C

LINEAR ALGEBRA • UNIT 4.01

# Determinants & Geometric Volume

PRACTICE WORKSHEET

25 Structured Exam-Style Questions  **Name:** _____ **Date:** _____ **Score:** _____ / 25

## Determinants & Geometric Volume Practice Exercises

**1** MCQ

Find $\|\mathbf{u}\|$ for $\mathbf{u} = (-9, -12)$.

A 21

B 14

C 15

D 225

**2** MCQ

A $3 \times 3$ symmetric matrix $Q_k$ has eigenvalues $\lambda_1 = 12$, $\lambda_2 = 3$, and $\lambda_3 = 14$. Determine the trace $\text{tr}(Q_k)$.

A 26

B 32

C 29

D 41

**3** MCQ

A linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, find $T(\mathbf{v})$.

A $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$

B $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$

C $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$

D $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

**4** MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If a point $P = (x, y)$ is transformed by $A$ to $P' = (x', y')$, such that $\begin{pmatrix} x' \\ y' \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}$, what is the image of the point $(3, 1)$ under this transformation?

A (7, 6)

B (6, 7)

C (8, 5)

D (5, 8)

**5** MCQ

Consider two vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

A 1

B 0

C -7

D 25

**6** MCQ

Consider the following system of linear equations represented by the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

. Use row reduction to transform the matrix into row-echelon form and determine the unique solution $(x, y)$ for the system. What is the value of $x$?

A 1

B 2

C 3

D 4

**7** MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$. Calculate the magnitude of $\mathbf{v}$, denoted as $\|\mathbf{v}\|$.

A 13

B 17

C 7

D 169

**8** MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$. If this vector is plotted on a standard $xy$-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?

A The vector's direction in the xy-plane.

B The angle the vector makes with the positive x-axis.

C The distance of the point (3,4) from the origin.

D The sum of the components of the vector.

**9** MCQ

Consider a parallelepiped in $\mathrm{R}^3$ formed by the origin and the three vectors $\mathbf{u} = (2\ 1\ k)$, $\mathbf{v} = (1\ 3\ 0)$, and $\mathbf{w} = (0\ 2\ 4)$. The volume of this parallelepiped is given as 24 cubic units. Find the value of $k$ if $k > 0$.

A 3

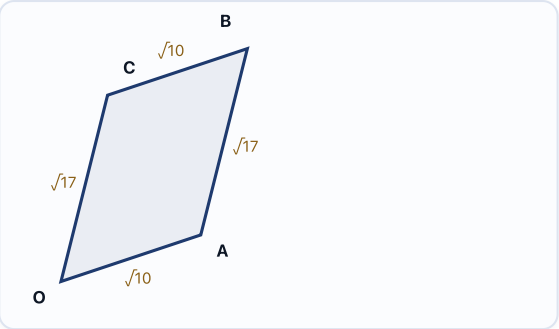
B 2

C 4

D 6

**10** MCQ

A parallelogram in the xy-plane is defined by the vectors $\mathbf{u} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$. What is the area of this parallelogram?




A 10

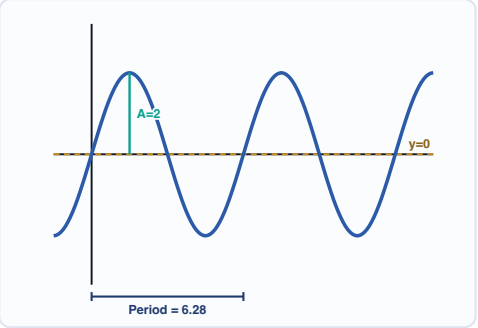
B 13

C 14

D 11

**11** MCQ

A solid object in $\mathrm{R}^3$ is defined by the region bounded by the planes $x = 0$, $y = 0$, $z = 0$, and the plane passing through the points $P = (6, 0, 0)$, $Q = (0, 4, 0)$, and $R = (0, 0, 3)$. If a linear transformation $T : \mathrm{R}^3 \to \mathrm{R}^3$ is represented by the matrix $M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$, what is the volume of the transformed solid object?




A 24

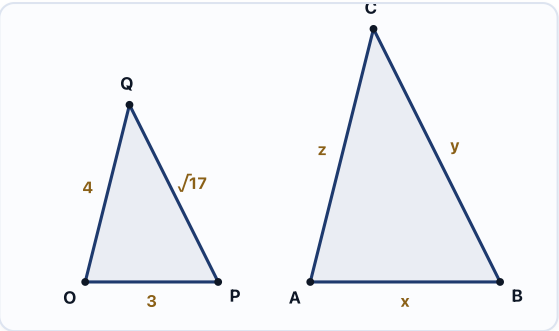
B 12

C 36

D 48

**12** MCQ

A parallelogram in the $xy$-plane is defined by the vertices $(0, 0)$, $(3, 1)$, $(1, 4)$, and $(4, 5)$. What is the area of this parallelogram?




A 13

B 15

C 11

D 17

**13** MCQ

Consider the parallelepiped in $\mathrm{R}^3$ defined by the vectors $\mathbf{u} = (2, 1, -1)$, $\mathbf{v} = (1, k, 2)$ with $k > 3$, and $\mathbf{w} = (3, 4, 1)$. If the volume of this parallelepiped is $\frac{45}{2}$ cubic units, find the value of $k$.

Given Vector Components

| Component x | Component y | Component z |
|---|---|---|
| 2 | 1 | -1 |

A $\frac{19}{2}$

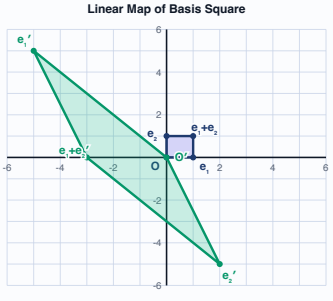
B $\frac{15}{2}$

C $\frac{13}{2}$

D $\frac{17}{2}$

**14** MCQ

For $A = \begin{bmatrix} -5 & 2 \\ 5 & -5 \end{bmatrix}$, find entry $(2A)_{22}$.




A -10

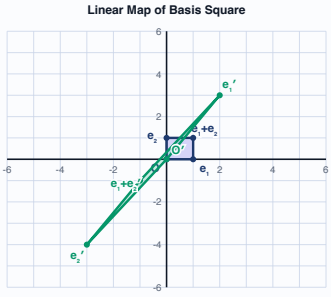
B -5

C 10

D -9

**15** MCQ

For $A = \begin{bmatrix} 2 & -3 \\ 3 & -4 \end{bmatrix}$, find entry $(-4A)_{21}$.




A 3

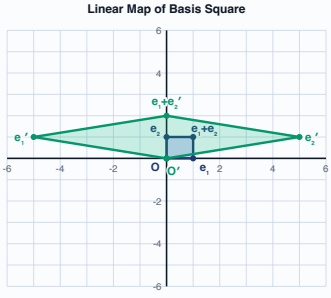
B 12

C -12

D -11

**16** MCQ

For $A = \begin{bmatrix} -5 & 5 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -6 \\ -2 & -4 \end{bmatrix}$, find entry $(A+B)_{11}$.




A -8

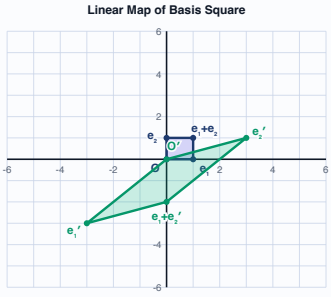
B -5

C -1

D -6

**17** MCQ

For $A = \begin{bmatrix} -3 & 3 \\ -3 & 1 \end{bmatrix}$, find entry $(4A)_{22}$.




A 5

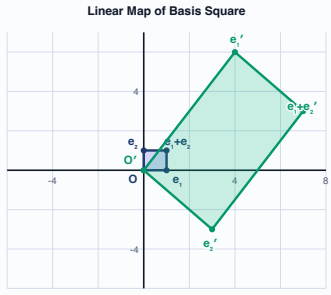
B 1

C -4

D 4

**18** MCQ

For $A = \begin{bmatrix} 4 & 3 \\ 6 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 5 & -6 \\ 5 & 4 \end{bmatrix}$, find entry $(A+B)_{11}$.




A 4

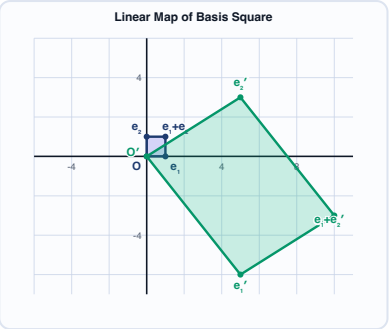
B 9

C 5

D 10

**19** MCQ

For $A = \begin{bmatrix} 5 & 5 \\ -6 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -4 & 5 \\ -5 & -5 \end{bmatrix}$, find entry $(A+B)_{21}$.




A -10

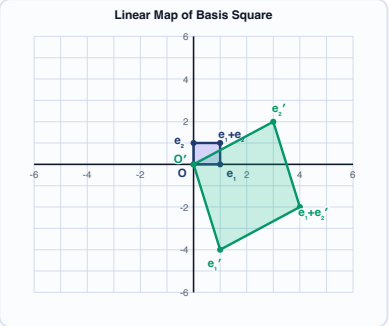
B -11

C -6

D -5

**20** MCQ

For $A = \begin{bmatrix} 1 & 3 \\ -4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 \\ -2 & -2 \end{bmatrix}$, find row 1 of $AB$.




A [7, -7]

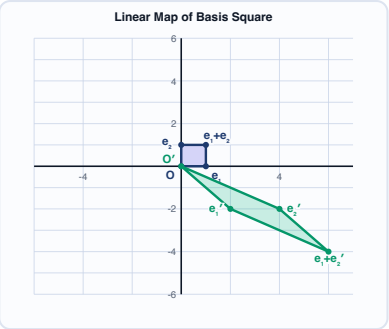
B [-8, -7]

C [-1, -6]

D [-7, -8]

**21** MCQ

For $A = \begin{bmatrix} 2 & 4 \\ -2 & -2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, find row 2 of $AB$.




A [0, 0]

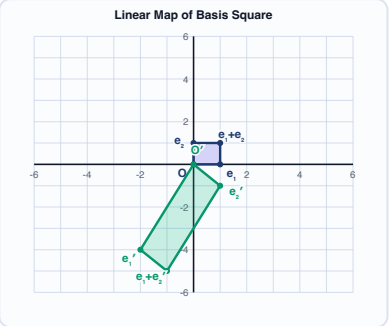
B [-4, -6]

C [-2, -2]

D [-2, 0]

**22** MCQ

For $A = \begin{bmatrix} -2 & 1 \\ -4 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 \\ -4 & 2 \end{bmatrix}$, find row 2 of $AB$.




A [0, -6]

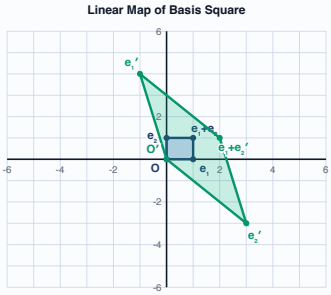
B [0, -18]

C [-18, 0]

D [-4, -2]

**23** MCQ

For $A = \begin{bmatrix} -1 & 3 \\ 4 & -3 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 \\ 2 & -2 \end{bmatrix}$, find row 2 of $AB$.




A [-10, 12]

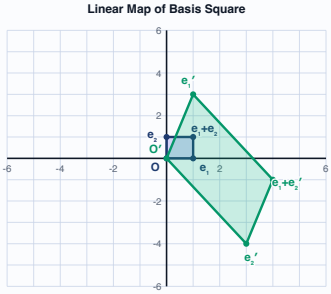
B [-2, -10]

C [-10, -2]

D [-4, 6]

**24** MCQ

For $A = \begin{bmatrix} 1 & 3 \\ 3 & -4 \end{bmatrix}$, $B = \begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix}$, find row 2 of $AB$.




A [2, -7]

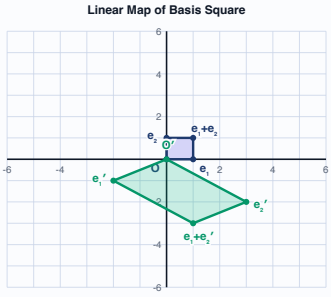
B [-2, -7]

C [-7, -2]

D [-6, -4]

**25** MCQ

For $A = \begin{bmatrix} -2 & 3 \\ -1 & -2 \end{bmatrix}$, $B = \begin{bmatrix} -4 & -1 \\ -2 & 3 \end{bmatrix}$, find row 2 of $AB$.




A [1, -12]

B [8, -5]

C [-5, 8]

D [4, -6]